



2.4 PRACTICAL EXAMPLES

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Several practical examples using capacitors are shown in figure 2.6. A $5\mu\text{F}$ electrolytic capacitor is used for DC blocking. It allows the signal to pass from one stage to the next while prevent the DC on one stage from being passed to the next stage. This occurs because the capacitor acts like a resistor of very low resistance for the signals and as a resistor of high resistance for DC.

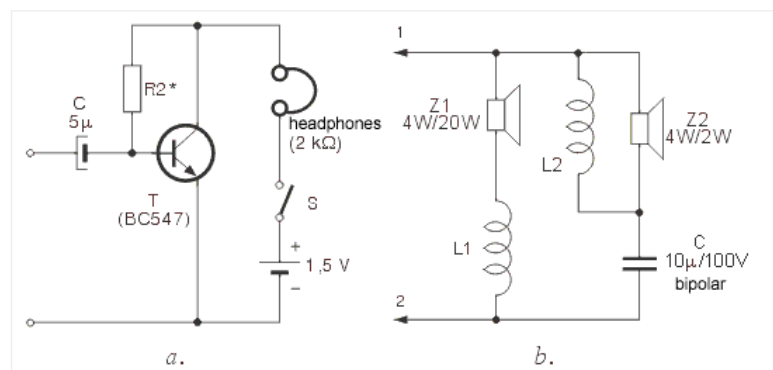


Fig. 2.6: a. Amplifier with headphones, b. Electrical band-switch

The figure 2.6b represents a diagram of a band-switch with two speakers, with Z1 used for reproducing low and mid-frequency signals, and Z2 for high frequency signals. 1 and 2 are connected to the audio amplifier output. Coils L1 and L2 and the capacitor C ensure that low and mid-frequency currents flow to the speaker Z1, while high frequency currents flow to Z2. How this works exactly ? In the case of a high frequency current, it can flow through either Z1 and L1 or Z2 and C. Since the frequency is high, impedance (resistance) of the coils are high, while the capacitor's reactance is low. It is clear that in this case, current will flow through Z2. In similar fashion, in case of low-frequency signals, current will flow through Z1, due to high capacitor reactance and low coil impedance.

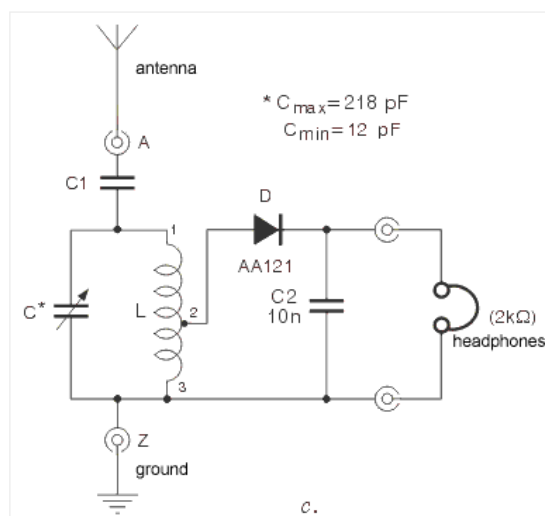


Fig. 2.6: c. Detector radio-receiver

The figure 2.6c represents a circuit diagram for a simple detector radio-receiver (commonly called a “crystal set”), where the variable capacitor C , forming the oscillatory circuit with the coil L , is used for frequency tuning. Turning the capacitor’s rotor changes the resonating frequency of the circuit, and when matching a certain radio frequency, the station can be heard.



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